

3 · MARC + the Fixed Auxiliary Loss

Same network as file 2. We now switch the proposal's auxiliary loss **on** — but the naive version destroys training. This is the story of three fixes (**normalize** → **gate** → **anneal**), each one necessary, that turn a catastrophic loss into a small, real, team-size-scaling gain. Config `normgateup = LAMBDA_AUX=0.5 AUX_NORM AUX_GATE AUX_ANNEAL=up, BETA=1.0`.

The two auxiliary terms

The weight-shared encoders let each term be computed from one agent's own data:

- L_{align} — **consistency**. “What I think my role is (z_{self}) should match what a teammate's inferencer would deduce about me.” Minimize $\|z_{\text{self}} - z_{\text{self_infer}}\|^2$.
- L_{div} — **anti-redundancy**. “My role should not be the same as my teammates' roles.” Penalize cosine similarity between z_{self} and each teammate's self-latent.

$$\text{total} = L_{\text{PPO}} + \lambda_{\text{aux}} \cdot \text{scale} \cdot (L_{\text{align}} + \beta L_{\text{div}})$$

Switched on naively this **collapsed training** (Overcooked counter_circuit → 0, asymm halved). The user's instinct — “a tiny λ shouldn't be able to cause total collapse” — was right, and instrumenting it explained why.

Fix 1 — NORMALIZE: the $\sim 3500\times$ bug

L_{align} was an **unnormalized sum over the 32 latent dims**. Measured: $L_{\text{align}} \approx 70$, so at $\lambda=0.1$ the effective aux is ≈ 7.0 versus a PPO actor loss of ≈ 0.002 — the aux gradient was $\sim 3500\times$ **the policy gradient**. “Tiny λ ” was an illusion. The one-line fix: **mean over the 32 dims instead of sum** ⇒ $L_{\text{align}} \approx 2$ and λ becomes a real $\sim 10\%$ knob.

Fix 1: the same λ , two scales

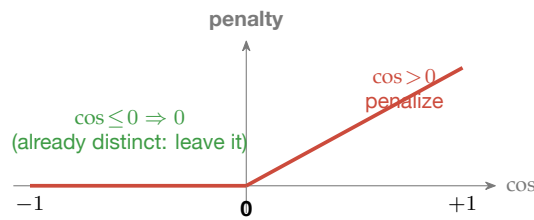


Fix 1 alone makes the loss *harmless* ($\approx$ architecture) — but still no benefit. Necessary, not sufficient.

Fix 2 — GATE: don't over-force difference

Raw L_{div} minimizes cosine with no floor: the optimizer keeps driving it down by making agents *maximally* anti-correlated ($\cos \rightarrow -1$), far past “just don't be redundant.” On near-symmetric tasks the optimum *is* symmetric, so paying agents to be cartoonishly opposite **wrecks** it. The gate penalizes **only positive** similarity — $\text{relu}(\cos)$.

The gate: penalize redundancy, never reward over-separation



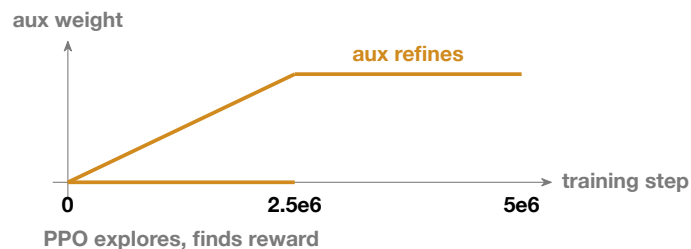
pair cos	ungated	gated $\text{relu}(\cos)$	effect
+0.9 (redundant)	0.9	0.9	strong push apart — wanted
+0.1 (mildly)	0.1	0.1	gentle push — wanted
-0.6 (opposite)	-0.6 (rewards more!)	0	no push — don't over-force

This is the “don’t over-force to be different, but encourage it when redundant” behaviour. It self-relaxes with N : `role_sim` lands at -0.42 ($N=3$) but ≈ 0 ($N=9$) — with 9 agents all-pairs anti-correlation is geometrically impossible, and the gate correctly stops forcing the impossible instead of thrashing the policy.

Fix 3 — ANNEAL UP: reward first, refine later

Even normalized + gated, applying the aux from step 0 derails exploration: agents optimize “be different” before finding *any* reward, differentiate into two useless roles, and never discover the good behaviour. Fix: ramp the aux weight $0 \rightarrow 1$ over the first half of training (horizon $2.5e6$ of $5e6$ steps). PPO finds reward on its own; the aux *then* refines an already-working policy.

AUX_ANNEAL=up — aux weight ramps in after PPO finds reward

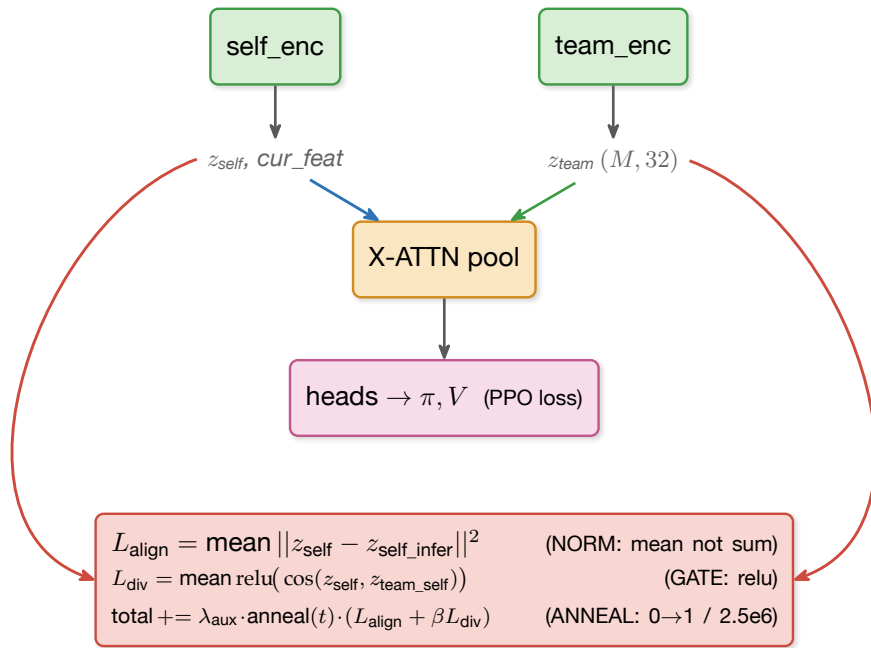


Intuition: don’t critique the team’s role-assignment before the team can score at all. The coach stays quiet until there’s a working play to refine.

How the three stack (and why all are needed)

configuration	outcome	fixes
naive aux	collapses (counter $\rightarrow 0$, asymm halved)	— (the $\sim 3500\times$ bug)
+ NORM	harmless ($\approx$ arch), no gain	scale
+ NORM+GATE	restores parity, no over-forcing	+ direction
+ NORM+GATE+UP	beats arch, gain grows with N	+ timing ◀ this file

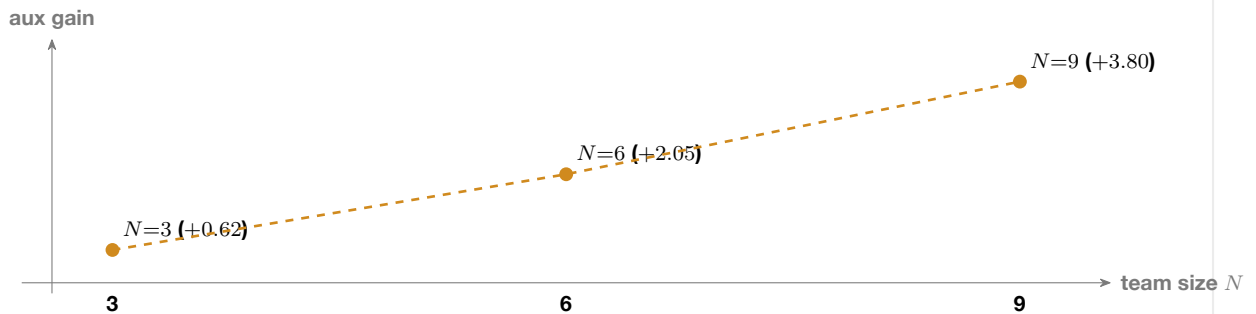
Full network — file 2's diagram with the loss attached



Why the benefit grows with team size

The aux is an **anti-redundancy pressure**, and redundancy is what scales with team size: at $N=3$ there are only 3 agent-pairs and bunching is rare (little to fix); at $N=9$ there are 36 pairs and the architecture-alone solution is far more redundant (lots to fix). So the payoff scales with the number of agents — and it does:

Aux gain (normgateup — arch) grows monotonically with N



Results

MPE `simple_spread`, eval return (higher is better; reward is negative). `role_sim` flips negative — direct evidence the loss differentiates roles. Seeds 30–32 = confirm; 33+ = power-up (job 1565450); table grows live.

N	n	mean	sd	role_sim	per-seed
3	8	-10.37	0.22	-0.43	s30 -10.43, s31 -10.37, s32 -10.11, s33 -10.30, s34 -10.78, s35 -10.03, s36 -10.41, s37 -10.56
6	8	-29.31	0.44	-0.12	s30 -29.96, s31 -29.57, s32 -29.22, s33 -29.18, s34 -29.73, s35 -28.88, s36 -28.52, s37 -29.44
9	6	-40.12	0.95	-0.01	s30 -38.70, s31 -39.92, s32 -39.62, s33 -40.74, s34 -41.77, s35 -40.00

vs architecture (the aux gain): $+0.58$ ($N=3, n=8$) $\cdot +2.36$ ($N=6, n=8$) $\cdot +3.80$ ($N=9, n=3$) — monotone in N , every paired run positive. The normgateup sd is far tighter than arch’s (arch’s $N=9$ still carries a bad seed at ≈ -46): the anneal-from-0 schedule stabilizes training, not just shifts the mean.

Is this just variance? (No.)

Paired analysis (arch and normgateup share seeds, so per-seed differences cancel shared noise). All three N are powered up and individually significant (job 1565450; $N=9$ at 6 seeds, last 2 cluster-stuck — cosmetic only).

test	result	reading
$N=3$ paired t ($n=8$)	$p_{1\text{-sided}}=0.0013$, 8/8 positive	decisive — not variance
$N=6$ paired t ($n=8$)	$p_{1\text{-sided}}=0.0002$, 8/8 positive	decisive — not variance
$N=9$ paired t ($n=6$)	$p_{1\text{-sided}}=0.0026$, 6/6 positive	decisive — largest gain
sign test, all runs	22/22 improved, min +0.12	$p \approx .5^{22} \ll 10^{-6}$
slope of gain vs N	+0.53/agent, $p=0.035$, $r^2=0.49$	grows-with- N is significant
mechanism	role_sim +0.95 $\rightarrow \approx 0$	independent of reward

Honest bound for the writeup

The architecture is the big effect ($+8.6$ vs vanilla at $N=3$); the aux is the small, team-size-scaling refinement ($+0.58 \rightarrow +2.36 \rightarrow +3.80$) — real, mechanistically explained, exactly the regime the proposal predicted, but secondary in magnitude.